Project Proposal
Deepfake Voice Detection

Introduction

With the rapid development of artificial intelligence and text-to-speech technology, it has become much easier to generate synthetic voices that sound very similar to real human speech. While these technologies can be useful for things like virtual assistants or accessibility tools, they can also create problems such as impersonation, misinformation, and identity fraud. Because of this, being able to detect whether a voice recording is real or AI-generated is becoming an important problem in machine learning and cybersecurity.

The purpose of this project is to explore both traditional machine learning and deep learning methods to classify voice recordings as either real human speech or synthetic speech. By building these models, we hope to create a system that can automatically detect AI-generated voices.

We will be using the Fake-or-Real (FoR) dataset from Kaggle, which contains over 195,000 audio recordings of both real human speech and computer-generated speech. The dataset combines several speech sources such as the Arctic dataset, LJSpeech dataset, and VoxForge dataset, along with synthetic speech produced using modern text-to-speech systems like Deep Voice 3 and Google WaveNet. Each recording is labeled as either real or synthetic, allowing models to learn patterns that distinguish between human speech and AI-generated voices.

Dataset link:

https://www.kaggle.com/datasets/mohammedabdeldayem/the-fake-or-real-dataset

Motivation:

Detecting synthetic speech is becoming more important as AI voice generation improves. Many modern technologies can produce voices that sound very realistic, which makes it difficult for people to tell whether a recording is real or fake. This creates risks in areas such as online scams, impersonation, and the spread of misleading information.

By using machine learning to analyze speech patterns, a system can automatically identify synthetic voices and help improve digital security. Tools like this could be useful for verifying audio recordings, preventing fraud, and supporting systems that rely on voice authentication.

Evaluation:

To evaluate the performance of our models, we will compare how accurately they classify voice recordings as real or synthetic. The dataset will be divided into training and testing sets so the models can be evaluated on unseen data.

Before training the models, the audio recordings will be converted into numerical data that the models can analyze and capture important characteristics of speech signals.

Several models will be implemented and compared in this project. A Convolutional Neural Network (CNN) will be used as the primary deep learning model because CNNs can learn patterns from audio features or spectrogram representations. In addition to the CNN model, traditional machine learning models such as Support Vector Machines (SVM) and Random Forest classifiers will also be tested.

To measure model performance, we will use metrics such as:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

These metrics will help us understand how well each model detects synthetic speech and whether more complex models like CNNs provide better results than traditional machine learning approaches.

Resources:
To complete this project, we will use the Fake-or-Real dataset from Kaggle, which contains over 195,000 labeled audio recordings of both real human speech and synthetic speech. This dataset will be used to train and evaluate the machine learning models.

For implementation, we will use Python because it provides strong support for machine learning and data analysis. Libraries such as NumPy and Pandas will be used for data preprocessing and organization. Scikit-learn will be used to implement machine learning models such as Support Vector Machines and Random Forest classifiers. For the deep learning model, we will use TensorFlow to design and train the Convolutional Neural Network.

We will be working on the project together using Google Colab.

Additional or Alternative Methods:

- Pretrain weights/parameters by applying the CNN made from scratch to a medium-sized balanced subset of the avspoof19 dataset [real size = 100,000+recordings] (will have around 5,000 audio clips, 2500 real, 2500 fake; split the fake set into 6 different attack types to balance the types of fake audio generated for the avspoof19 detection competition [there are also 11 unknown attack types that could be separated into classes by some type of machine learning and defined as unique subsets to increase performance])
- Model Setup

a. (optional – denoising or data augmentation on Fake or Real medium-sized balanced subset [real size = 195,000+ recordings] of 10,000 recordings — 5000 real = 5000 fake — using RawBoost)

1. Use a feature extractor layers, AKA filter, like Hubert, Wavelet(s), Whisper, on the raw audio
2. Convert those extracted features into mel spectrogram images
3. Process MS images using our specific CNN from scratch (based on the powerpoint from this class), and other models for comparative analysis, to classify the deepfake audio as real or fake.
    a. CNN variables for experimentation
        i. Learning rate
        ii. Optimizers
        iii. Number of layers or batch size
        iv. Size of filters
        v. Stride

The evaluation metrics that would be useful for this project include balanced accuracy, loss, ROC, and EER. Balanced accuracy should be tracked instead of standard accuracy because standard accuracy can provide misleading high scores to models that correctly classify imbalanced datasets. Imbalanced datasets do not generalize well to other datasets gathered to reflect variety in the real-world. Graphs can be created to track the changes in balanced accuracy and loss over epochs for each model. An ROC curve graphic could illustrate the tradeoff between true positives and false positives (a steeper curve would show that the model successfully separated real and fake classes, correctly classifying fakes instead of mistaking fakes for real audio). Finally, an EER measurement table could be added to show the industry standard for measurements of deepfake detector performance. EER captures False Positive (false alarms) and False Negative (misses) to show the percentage of wrong classifications made when there is an equal amount of either of these mistakes (ie FP = FN).

This project could compare results between the CNN made from scratch, YOLOv8, Rawformer, and the AUCO Resnet. These preset testing models produced low EERs on some of the benchmark datasets used in a 2024 paper written by Uliana Zbezhkhovska. Each model should be trained to 100 epochs, but if the model's/models' complexity creates time or processing constraints, the epochs could be lowered to a minimum of 50.

Additional Methods Works Cited:

Zbezhkhovska Uliana. On effectiveness and generalization capabilities of deep learning models for deepfake audio detection. Ukrainian Catholic University, Faculty of Applied Sciences, Department of Computer Sciences. Lviv 2024, 44 p.

Contributions:

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